

Idea Proposal

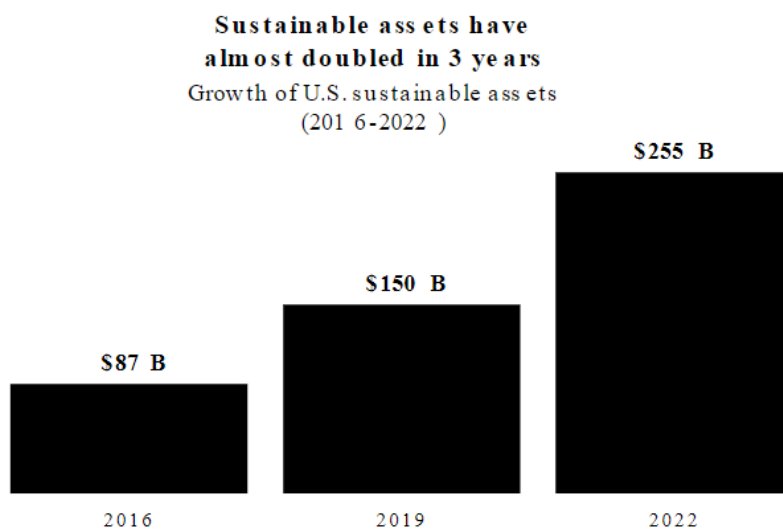
Climate change and renewable energy in the USA

University of Nevada Las Vegas

MIS 776: Business Intelligence

September 30, 2023

For my proposal, I have selected climate change and renewable energy for several reasons. It is a current and multifaceted issue, touching upon environmental, economic, political, social, and ethical aspects. From an environmental perspective, the current rate of warming is unprecedented in the last 10,000 years, as reported by NASA (NASA, 2023b). While the Sun has indeed played a role in global warming, there is insufficient evidence to claim it as the sole culprit (NASA, 2023c). This leads to economic considerations, such as the rising gas prices and the potential for investments in renewable energy resources. In fact, the growing demand from investors has significantly contributed to the expansion of sustainable funds, with such assets nearly doubling to \$255 billion in just three years, as illustrated in the graphic below (ESG investing & funds, 2023).



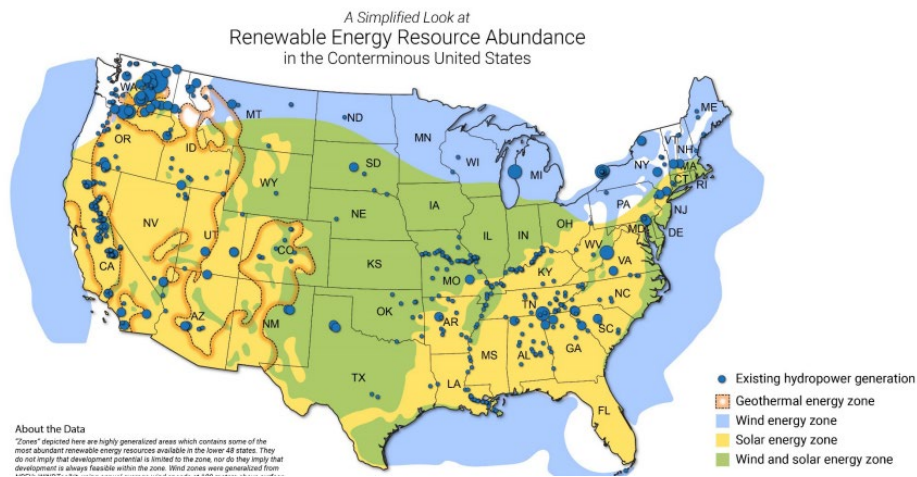
The renewable energy sector presents a substantial and growing market encompassing a lot of technologies, including but not limited to solar panels, wind turbines, hydroelectric power systems that harness the energy of flowing water through turbines to generate electricity,

geothermal heat pumps, and a multitude of other innovative solutions. In this context, conducting in-depth analyses to assess and improve the efficiency of these existing technologies could also benefit business and investors. This could not only save money to people that use sustainable energy but also make money to people that wants to invest in this sector. This study not only contributes to the continued growth of the renewable energy market but also play a crucial role in accelerating the transition towards sustainable and environmentally responsible energy sources.

Renewable energy holds significant potential from other perspectives. From a political standpoint, it can be linked to gas prices and incentives for renewable energy, potentially leading to a global transition toward greener resources. Ethical and social considerations involve conserving and protecting water resources through efficiency, reuse, and effective stormwater management, waste reduction, pollution prevention, increased recycling, as well as the design, construction, maintenance, and operation of high-performance sustainable buildings, among other factors (NASA, 2022). Moreover, every facet of renewable energy offers opportunities for more sustainable profit generation and presents new business and investment prospects.

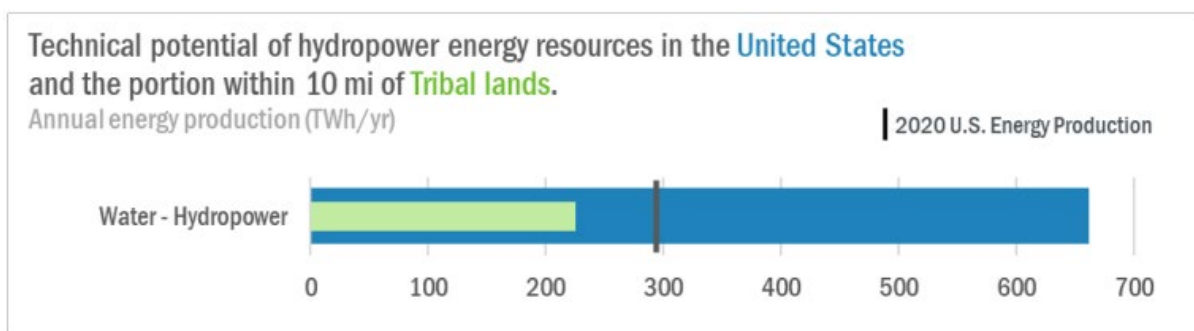
I have chosen this issue because I believe in environmental sustainability. Coming from a small city in northern Italy, where people rely on trains, bikes, and electric buses for transportation, I see tremendous potential in this sector worldwide, including the United States. Additionally, operating a business with a commitment to ethical and sustainable practices increases the likelihood of success. Furthermore, sustainable energy can pave the way for the emergence of new businesses and investment opportunities in the market. Understanding the required amounts of solar energy, wind energy, hydrogen energy, and geothermal energy is crucial for making informed decisions. Lastly, monitoring changes in the weather is essential to determine suitable locations for implementing sustainability initiatives.

Data can be effectively presented through various visualizations created using tools such as Tableau or Python. Here are some examples of these visualizations.



(U.S. Energy Information Administration – EIA)

This illustrates the suitability of different renewable energy sources for various regions of the United States. In my research, I aim to track how these zones are shifting over the years and possibly predict future trends. This knowledge can help optimize construction timing and guide strategic investments.



(U.S. Energy Information Administration – EIA)

Another intriguing visualization to consider is the potential for sustainable energy in the United States compared to our actual production. The U.S. Department of Energy gathers real

data on all sustainable energy and climate change factors, offering yearly, quarterly, and monthly datasets for analysis. Valuable insights can be gleaned from monthly data and heatmaps, shedding light on both monthly and yearly growth patterns. Techniques such as machine learning will aid in predicting future trends within the data. As far as data cleaning is concerned, I will utilize Python with libraries such as Pandas and NumPy. To present the research to investors, it would be ideal to create a report that covers all the key points and can be easily updated. I will focus more on visualization so that companies and investors can easily read the data and make the best decisions.

Resources

NASA. (2023b, September 27). *Climate change evidence: How do we know?*

<https://climate.nasa.gov/evidence/>

NASA. (2023c, September 27). *The causes of climate change*. NASA.

<https://climate.nasa.gov/causes/>

ESG investing & funds. BlackRock. (n.d.). [https://www.blackrock.com/us/financial-](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

[professionals/investment-](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

[strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3A](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

[google%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

[oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP5](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

[81tykaAmp2EALw_wcB&gclsrc=aw.ds](https://www.blackrock.com/us/financial-professionals/investment-strategies/sustainable?cid=ppc%3Auswa_us%3Auswa_index_nb_sustainability_exact%3Agoogle%3Anonbrand_nonprod%3Afa&gclid=Cj0KCQjwjt-oBhDKARIsABVRB0xx9TSfUyKNFhDNy5XLaul75dFFUG9szL5mKsHcYVfCOzGdP581tykaAmp2EALw_wcB&gclsrc=aw.ds)

NASA. (2022, July 18). *NASA Global Climate Change: Sustainability and Government Resources*.

NASA. <https://climate.nasa.gov/solutions/resources/>

U.S. Energy Information Administration - EIA - independent statistics and analysis. Where solar is found - U.S. Energy Information Administration (EIA). (2023, April).

<https://www.eia.gov/energyexplained/solar/where-solar-is-found.php>

Chat GTP spelling check.